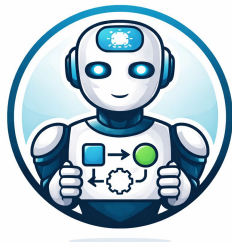


Agentic AI for Tech Professionals

Introduction to N8N



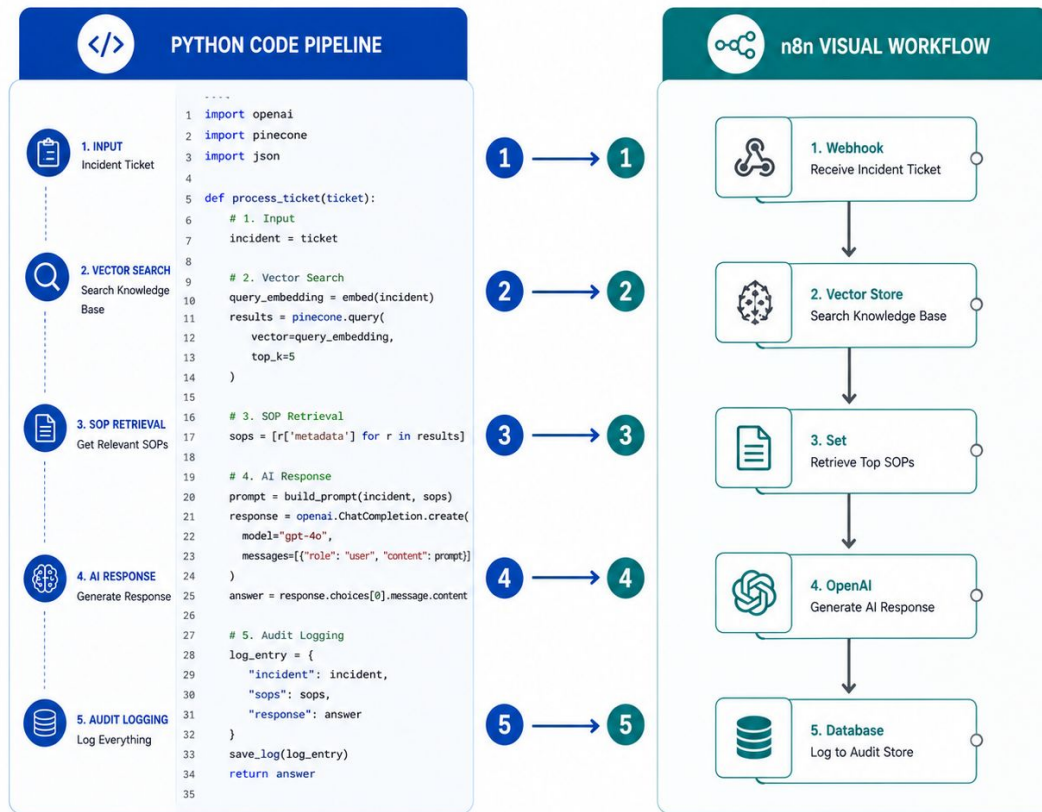
From Python Logic → Visual Workflows

n8n transforms code pipelines into visual automation systems

- Visual workflow automation platform
- Connect APIs, AI, databases
- Build pipelines without large codebase
- Perfect for orchestration
- Enterprise integration friendly

AUTOMATION PIPELINE: CODE vs VISUAL WORKFLOW

Same Logic. Different Approach.



Python Code

Full control, flexible,
developer-driven



Same Logic
Same steps,
same outcome



n8n Workflow

Visual, modular,
easy to modify

From Python Logic → Visual Workflows

Yesterday, we built:

- agents
- orchestrators
- routing logic
- RAG systems

Today:

👉 we convert those concepts into workflows.

We visually connect:

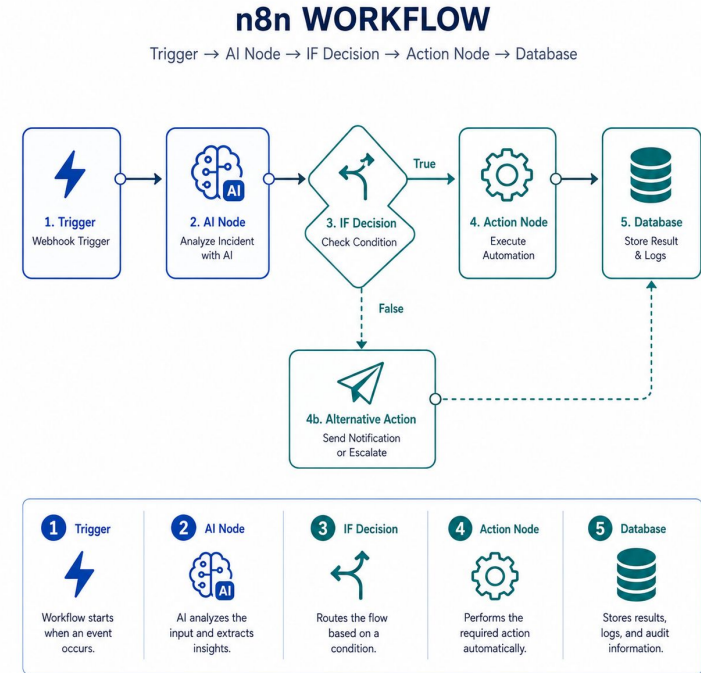
- triggers
- AI
- routing
- APIs
- storage

This is how many enterprise automation systems are built.

Core n8n Workflow Structure

Every workflow follows a pipeline pattern

- Trigger starts workflow
- Nodes process data
- IF nodes make decisions
- APIs extend capabilities
- Outputs trigger actions



Core n8n Workflow Structure (2)

n8n workflows are basically:

👉 visual orchestrators

Exactly like yesterday's Python orchestrator.

Pipeline:

Input

→ Process

→ Decide

→ Act

Core n8n Workflow Structure (3)

Example:

New incident ticket

→ classify with AI

→ assign priority

→ route to team

→ store in database

Same architecture.

Different implementation.

What is a Trigger?

A trigger starts the AI workflow automatically

- Entry point of automation workflow
- Starts workflow execution
- Can come from multiple sources
- Enables event-driven AI systems
- Common in enterprise automation

What is a Trigger? (2)

A trigger is:

👉 what starts the workflow.

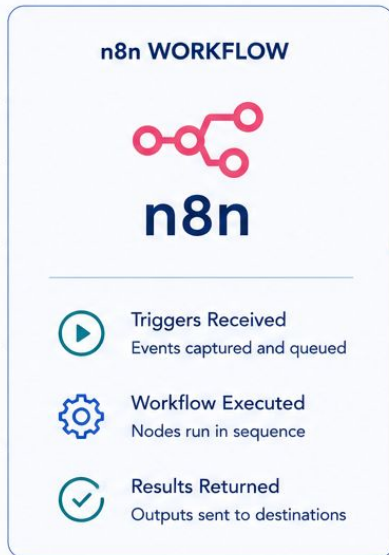
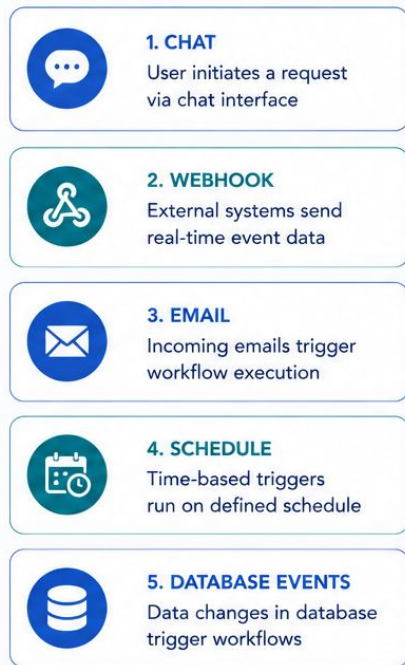
Examples:

- Chat message
- Webhook
- Email received
- Schedule
- Database event

n8n Workflow Trigger Sources

Multiple event sources. One intelligent workflow.

TRIGGER SOURCES



WORKFLOW OUTPUTS



HOW IT WORKS



What is a Chat Model?

The chat model is the reasoning engine of the AI agent

- Processes prompts and context
- Generates AI responses
- Different models fit different use cases
- Temperature controls randomness
- Model choice affects performance and cost

What is a Chat Model? (2)

The chat model is:

👉 the AI brain.

Examples:

- GPT-4o
- Claude
- Gemini

The model:

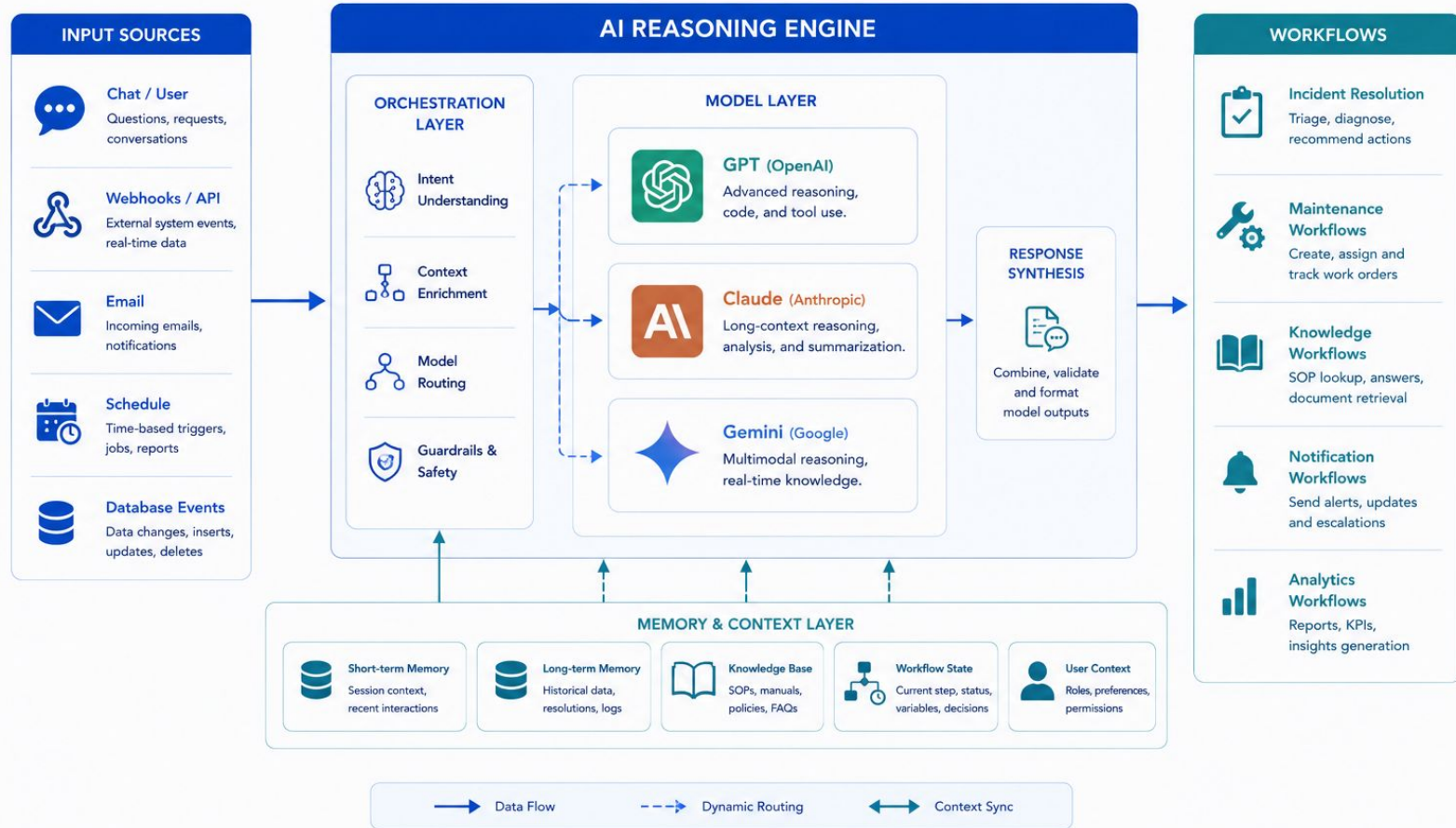
- understands input
- reasons about tasks
- generates responses

Different use cases require different models.

Use Case	Preferred Style
Email assistant	Fast + concise
Incident analysis	Structured + reliable
Decision workflow	Low randomness

AI REASONING ENGINE

Multiple AI models. Unified reasoning. Connected workflows.



What is a Tool?

Tools allow AI agents to interact with external systems and perform actions

- Extends AI capabilities
- Enables real operational actions
- Can connect APIs and databases
- Supports email and file systems
- Core to enterprise AI automation

What is a Tool? (2)

Without tools,

AI can only:

👉 generate text

Tools allow AI to:

- send emails
- call APIs
- retrieve files
- update databases
- trigger workflows

Examples in n8n:

- Gmail node
- HTTP Request node
- Google Drive node

This transforms AI from:

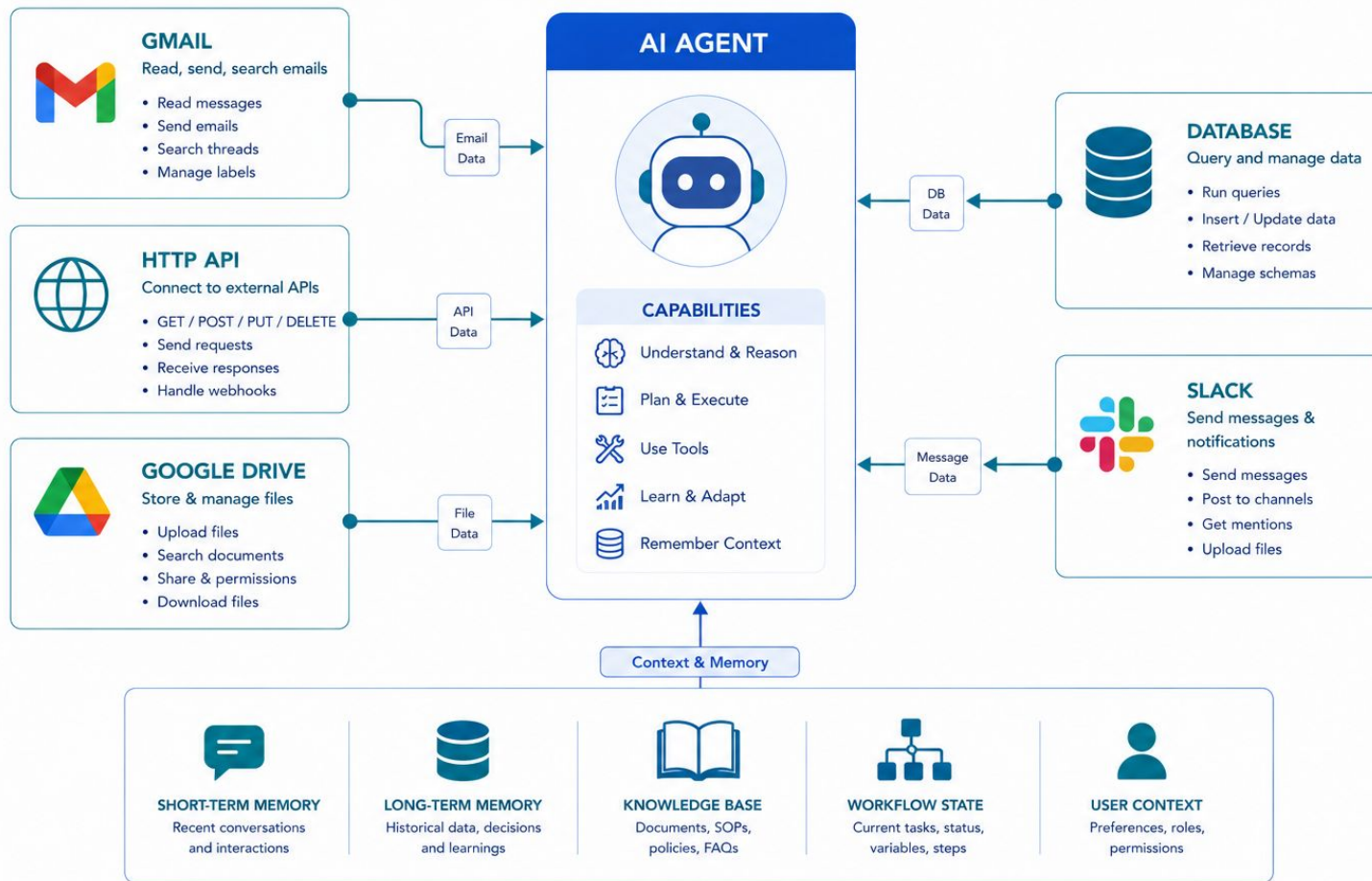
✗ assistant

to

✓ operational automation system.

AI Agent with Connected Tools

One agent. Multiple tools. Intelligent outcomes.



What is Memory?

Memory allows AI agents to retain context across interactions

- Stores conversation history
- Maintains workflow context
- Improves response continuity
- Supports long-running workflows
- Context window has limits

What is Memory? (2)

Without memory,
AI forgets previous interactions.

Example:

Without memory:

User: My name is Ali

User: What is my name?

AI: I don't know

With memory:

AI: Your name is Ali

Memory is critical for:

- conversational assistants
- workflow continuity
- operational systems

But memory also has limits:

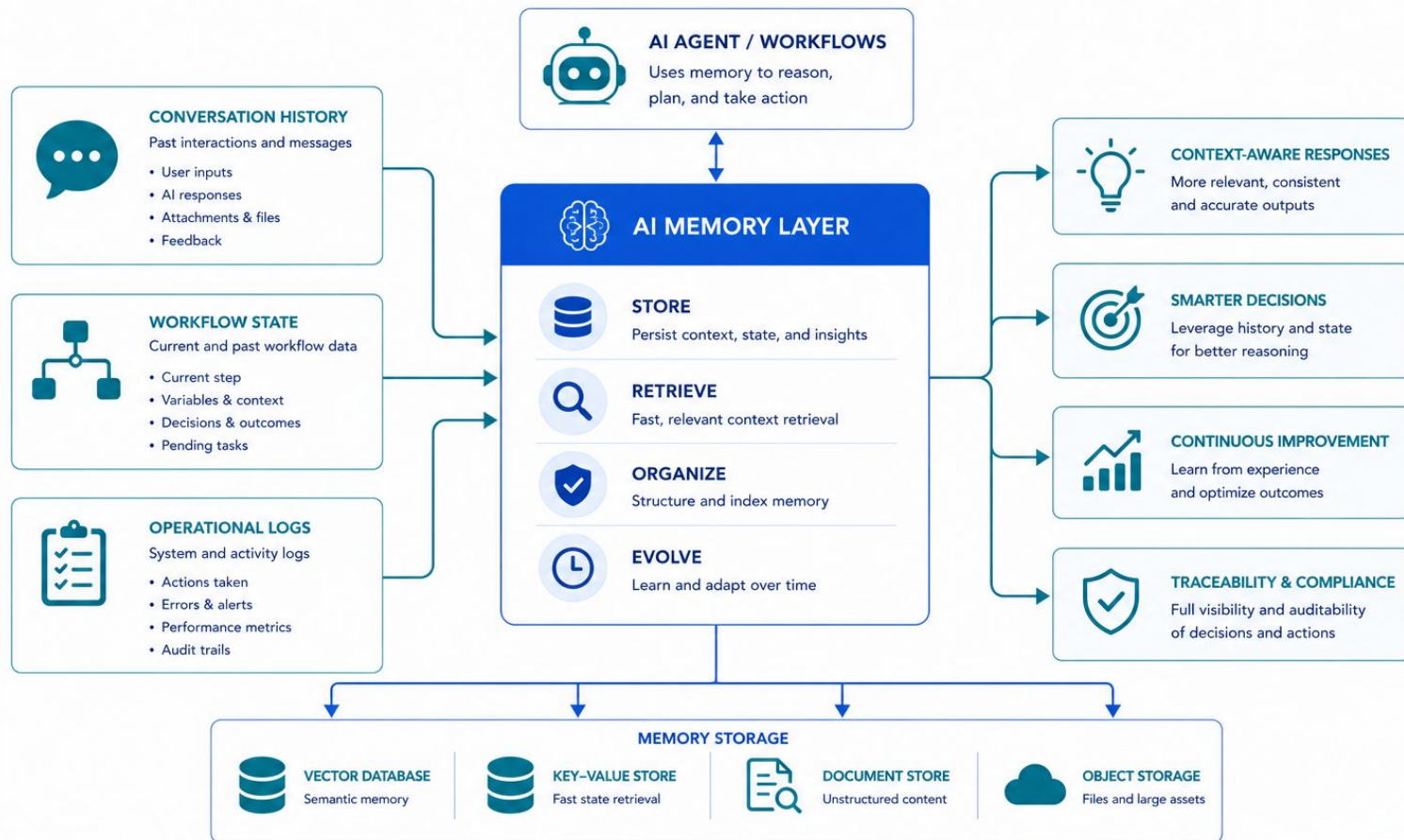
👉 context window length.

Too much memory can:

- increase token cost
- reduce performance
- introduce noise

AI MEMORY LAYER

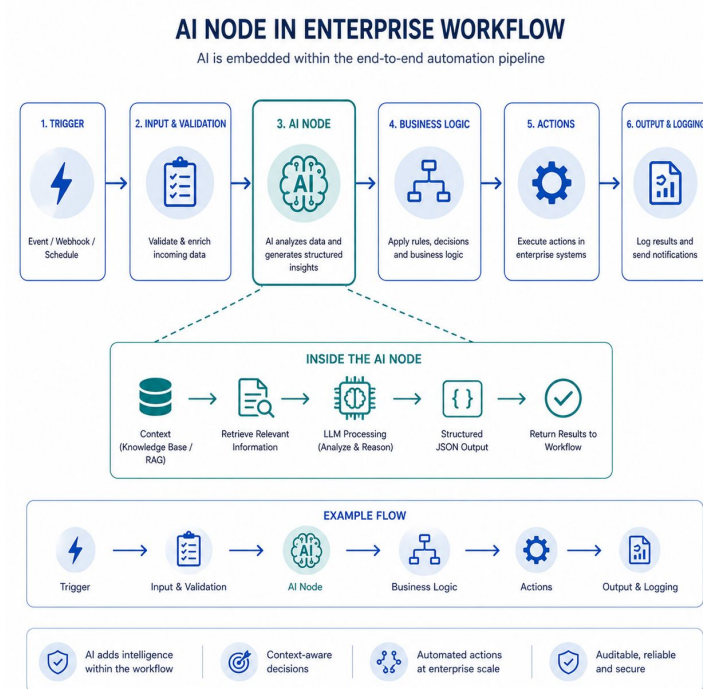
Persistent context. Smarter decisions. Better outcomes.



How AI Fits into Workflows

AI becomes one step inside a larger automation system

- AI node processes text
- Workflow controls execution
- AI output drives decisions
- Structured JSON preferred
- AI is NOT the entire system



How AI Fits into Workflows (2)

One common mistake:

People think:

👉 AI = entire application

Wrong.

AI is usually:

👉 ONE component inside a workflow.

The workflow still controls:

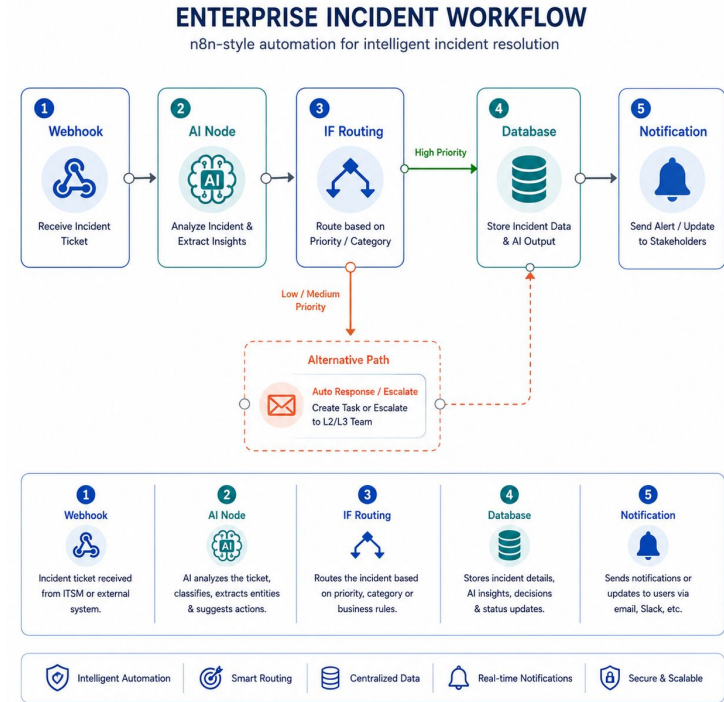
- routing
- approvals
- logging
- storage
- notifications

This is critical enterprise thinking.

Manufacturing Incident Workflow in n8n

We will rebuild the incident system visually

- Receive incident input
- Send to AI classifier
- Route by category
- Generate response
- Store result automatically



Today's goal

Take the Day 1 Python system and rebuild it visually in n8n.

By end of today:

participants will build:

- ✓ AI classification
- ✓ routing workflows
- ✓ automated storage
- ✓ enterprise automation pipelines

This becomes:

👉 production workflow thinking